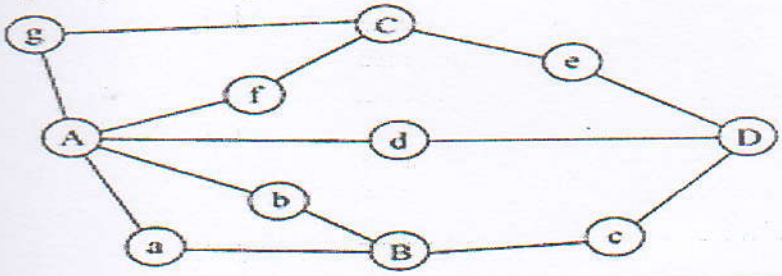
SOMAIYA VIDYAVIHAR UNIVERSITY

21·05·2025 (E)

Semester: Jan 2025– April 2025		
Maximum Marks: 50 Examination: End-Semester Examination Duration: 2 Hrs.		
Programme code: 01 Programme: BTech Computer Engineering	Class: SY	Semester: III (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department/Section/Center: COMP	
Course Code: 216U01C302	Name of the Course: Data Structures	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Define Data structure. List different operations performed on Data Structures.	05
ii)	Write a Pseudocode/algorithm called 'push' that takes two parameters: an integer variable and a stack into which it would push this element and returns a 1 or a 0 to show success of addition or failure.	05
iii)	Differentiate the linked list and linear array	05
Q.2	Attempt any **one**	10
i)	Explain a circular queue . Write pseudocode/algorithm to implement following operations on circular Queue. Consider all scenarios: i) Insertion ii) Deletion	
ii)	Explain the Set Abstract Data Type (Set ADT). How are disjoint sets implemented using trees in the Union-Find data structure? Illustrate your answer with a diagram or example.	
Q.3	Attempt any **one**	10
i)	Construct AVL Tree for the following sequence of numbers- 52 , 20 , 60 , 10 , 8 , 15 , 32 , 46 , 11 , 48	
ii)	Illustrate DFS and BFS traversals of following graph and also represent same using adjacency list and adjacency matrix  	
Q.4	Attempt the following	10
i)	Construct a Binary Search Tree (BST) for the following sequence of numbers- 50, 70, 60, 20, 90, 10, 40, 100 What is the in-order traversal of the final BST?	

	What is the pre-order traversal of the final BST? What is the post-order traversal of the final BST? What is the height of the BST after all insertions?	
Q.5	Attempt the following	10
i)	Given the input {4371, 1323, 6173, 4199, 4344, 9679, 1989}, a fixed table size of 10, and a hash function $H(X) = X \bmod 10$, show the resulting a. Linear probing hash table b. Separate chaining hash table	5
ii)	Sort the numbers 82, 73, 12, 39, 26, 88, 2, 9, 60 and 41 using shell sort.State the output after each pass	5